Basic Networking Skills for Digital Forensics

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Android Forensics Case Studies

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1. **Introduction**

   This report presents the forensic examination of a company-issued Google Pixel 3 (Android 10) submitted by Harbour Operations Ltd under case reference SBT-DF205-C1, following suspected unauthorised communication and sharing of information. The evidence is Joshua Hickman's Android 10 file system extraction (Pixel 3.zip, 5,247,820,897 bytes) from Digital Corpora. This part of the examination aims to preserve the evidence, establish the device identity and its users, and identify the carrier and account associations that later communication, media and location analysis depend on. Integrity was recorded with MD5 and SHA-256 hashes before analysis, and examination was carried out on a working copy using Kali Linux command-line tools and DB Browser for SQLite. All findings come from records in the supplied extraction only; no live accounts were accessed, and no passwords or tokens are reported. Raw timestamps are retained and normalised to UTC. This section covers acquisition and integrity, device properties, SIM and carrier information, user profiles and account associations.

2. **Evidence Acquisition and Integrity**

Field	Value
Original file Name	Pixel 3.zip
Source	digitalcorpora
File size	5,247,820,897 bytes (5.25GB)
Download date	26/10/2026
Md5	9cc37ebbbc4e918ee5427de1fe1deecc
Sha256	ca6918ef8b20486b6a5ded15609ac51318f37829480f93be3ba15364a8aa00a
Working Copy	Pixel 3.zip
Working copy (Hashdeep)	ca6918ef8b20486b6a5ded15609ac51318f37829480f93be3ba15364a8aa00a
note	Original copy not tempered

2.1Folder structure creation

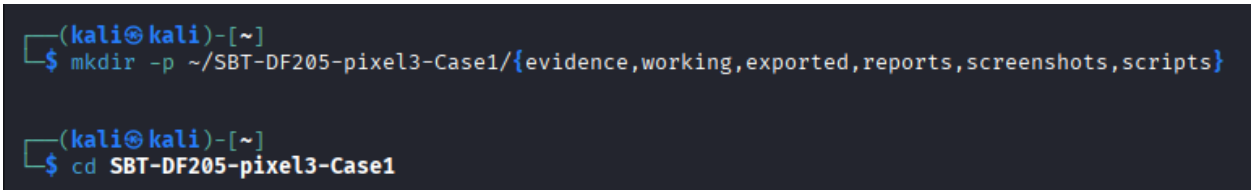


Figure 1. E01: workstation created with folders for evidence preservation and reports documentation.

2.2Verifying Integrity

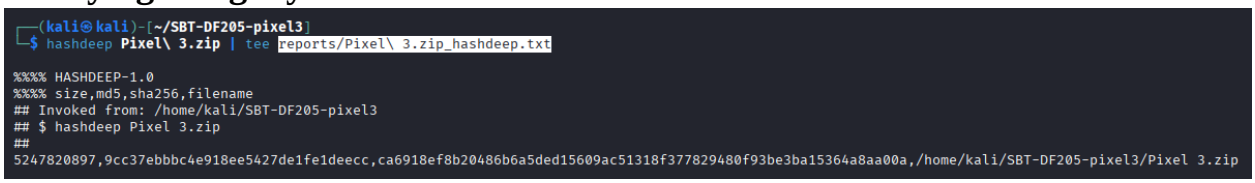


Figure 2. E02: hashdeep pixel/ 3.zip | tee reports/Pixel\ 3.zip_hashdeep.txt, recorded the integrity before any analysis

2.3Unzipping to get the pixel 3 image

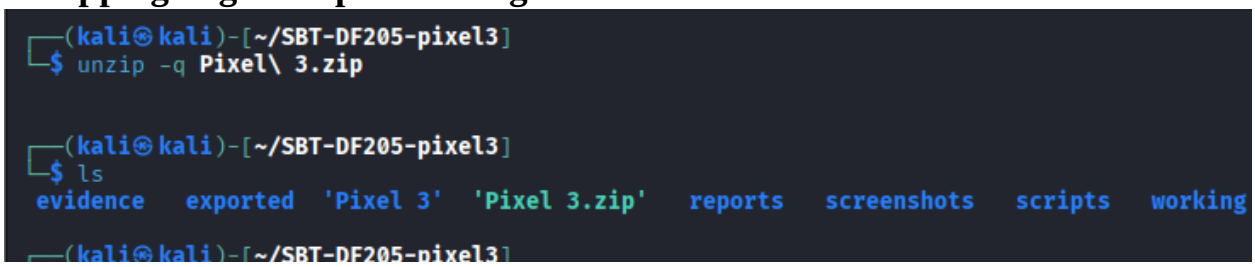


Figure 3. E0 3. File unzipped and image successfully obtained for further examination.

2.4Extracting the device information

```
┌──(kali㉿kali)-[~/SBT-DF205-pixel3]
└─$ cat 'Pixel 3/product/build.prop' | tee reports/build.prop.txt


# begin common build properties
# autogenerated by build/make/tools/buildinfo_common.sh
ro.product.build.date=Tue Dec 17 19:34:12 UTC 2019
ro.product.build.date.utc=1576611252
ro.product.build.fingerprint=google/blueline/blueline:10/QQ1A.200105.003/6079943:user/release-keys
ro.product.build.id=QQ1A.200105.003
ro.product.build.tags=release-keys
ro.product.build.type=user
ro.product.build.version.incremental=6079943
ro.product.build.version.release=10
ro.product.build.version.sdk=29
ro.product.product.brand=google
ro.product.product.device=blueline
ro.product.product.manufacturer=Google
ro.product.product.model=Pixel 3
ro.product.product.name=blueline
# end common build properties
```

```
┌──(kali㉿kali)-[~/SBT-DF205-pixel3]
└─$ grep patch  'Pixel 3/system/build.prop' | tee reports/security_patch.txt

ro.build.version.security_patch=2020-01-01
```

Figure 4. E04. Device building properties extracted using (build.prop), the device was built on Dec 17 19:34:12 UTC 2019 and the ROM was compiled on 1576611252 UTC, type: user, 10th version release, software development kit =29 and was manufactured by Google. Attached also is the security patch of the device

```
# ADDITIONAL PRODUCT PROPERTIES
#
ro.build.characteristics=nosdcard
persist.vendor.debug.sensors.accel_cal=1
masterclear.allow_retain_esim_profiles_after_fdr=true
ro.apk_verity.mode=1
```

Going through the additional product properties from the product information to check if the system has SD Cards. The device has no SD Cards attached to it.

3. Mobile Carrier information extraction

```
┌──(kali㉿kali)-[~/SBT-DF205-pixel3]
└─$ tree 'Pixel 3/data/user_de/0/com.android.providers.telephony/' | tee reports/telephony.txt

Pixel 3/data/user_de/0/com.android.providers.telephony/
├── databases
│   ├── carrierIdentification.db
│   ├── CarrierInformation.db
│   ├── cellbroadcasts.db
│   ├── HbpcdLookup.db
│   ├── mmssms.db
│   └── telephony.db
└── shared_prefs
    ├── build-id.xml
    ├── CarrierIdProvider.xml
    ├── preferred-apn.xml
    └── preferred-full-apn.xml

3 directories, 10 files
```

Figure 5. (E05) Examining all files under com.android.providers.telephony folder. The examination identified carrier related databases and configuration files within the device telephony provider directory. The files above establish presence of relevant carrier identification artifacts.

3.1 Tables listening on telephony.db

Initial triage

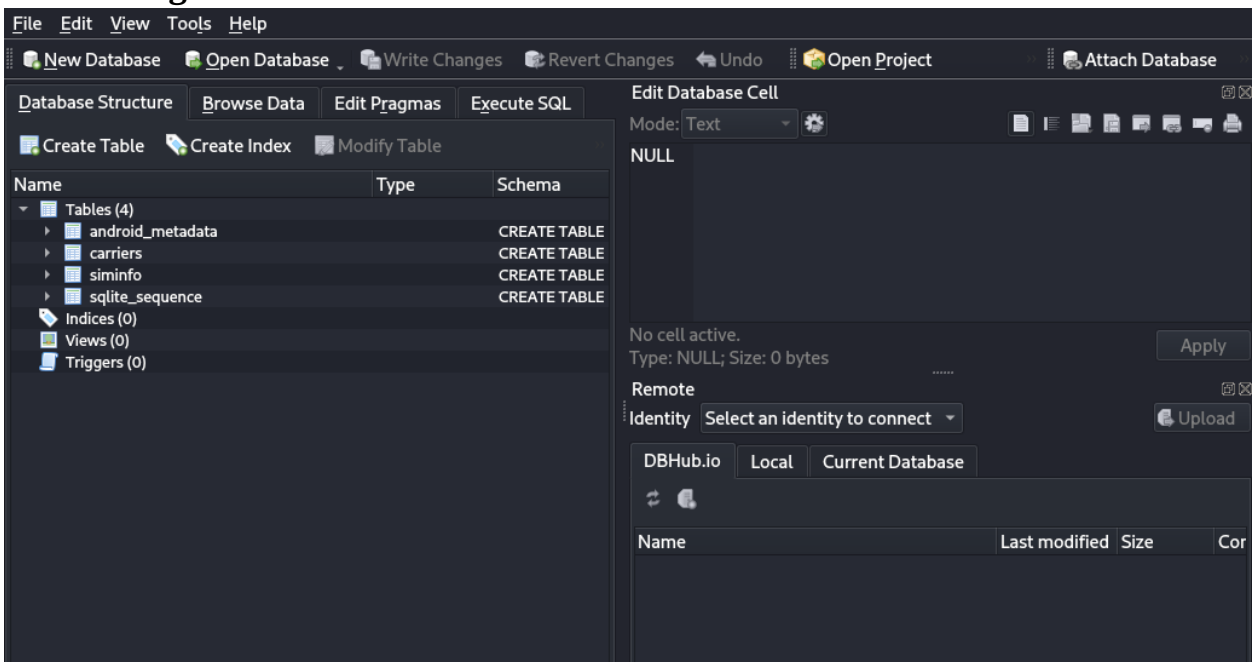


Figure 6. (sqlitebrowser 'Pixel 3/data/user_de/0/com.android.providers.telephony/databases/telephony.db').

3.2Sim Info.

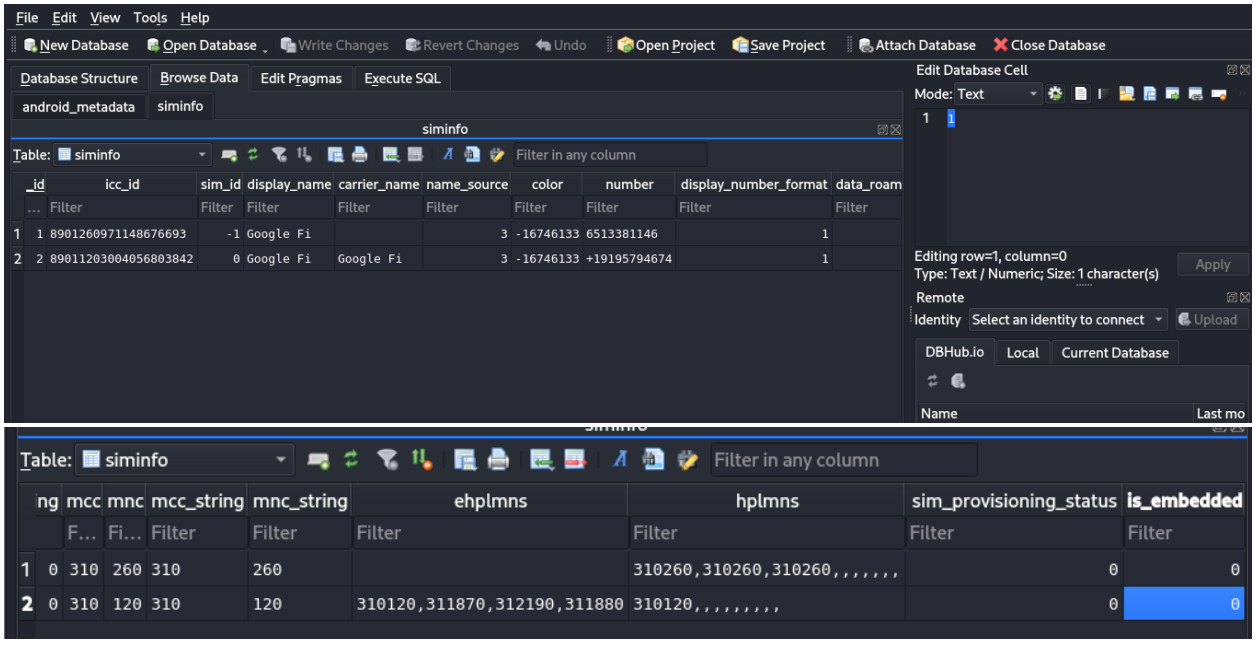


Figure 7. E06: this is the full display of the sim table information Obtained /extracted data's

ID	*Sim ID*	*Sim Serial Number (Icc ID)*	*Display name*	*Carrier name*	*number*	*Mobile Country Code (mcc)*	*M N C (*
1	*-1*	*8901260971148676693*	*Google Fi*	*NiL*	*6513381146*	*310*	*2*
2	*0*	*89011203004056803842*	*Google Fi*	*Google Fi*	*+19195794674*	*310*	*1*

The results from the sim examination table give out information that consists of sim serial numbers (which is usually found on the physical sim and can be traced), and the actual sim number also found in the sim table analysis.

Checking blocked numbers

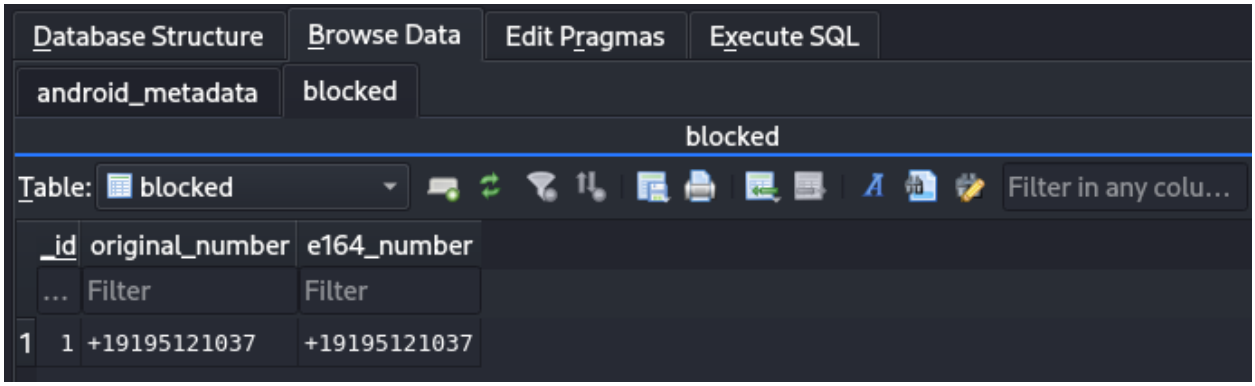


Figure 8. On the sim information, +19195121037 was found on the blocked contact list.

Checking carrierconfig information to verify the carriers

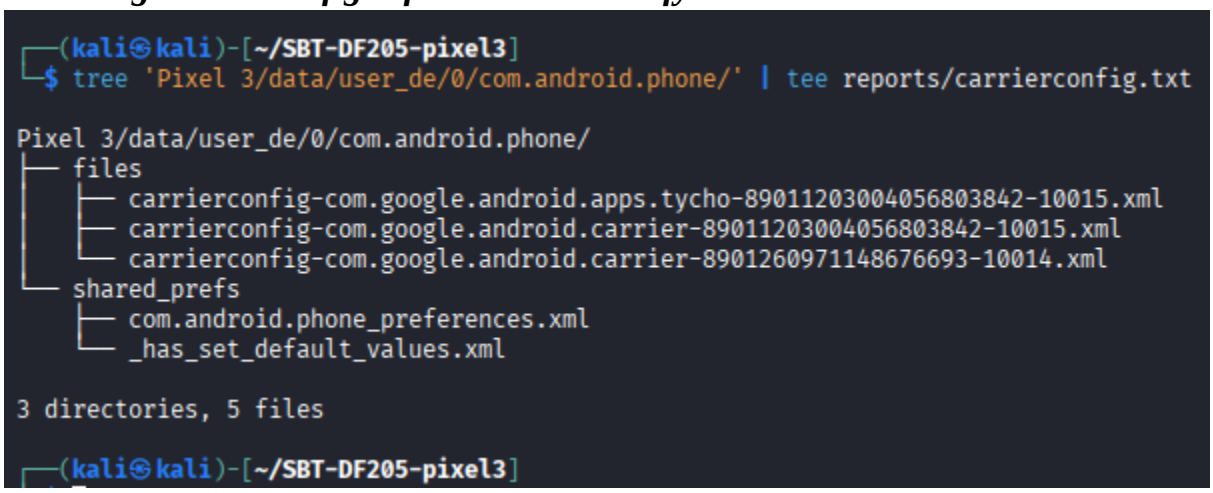
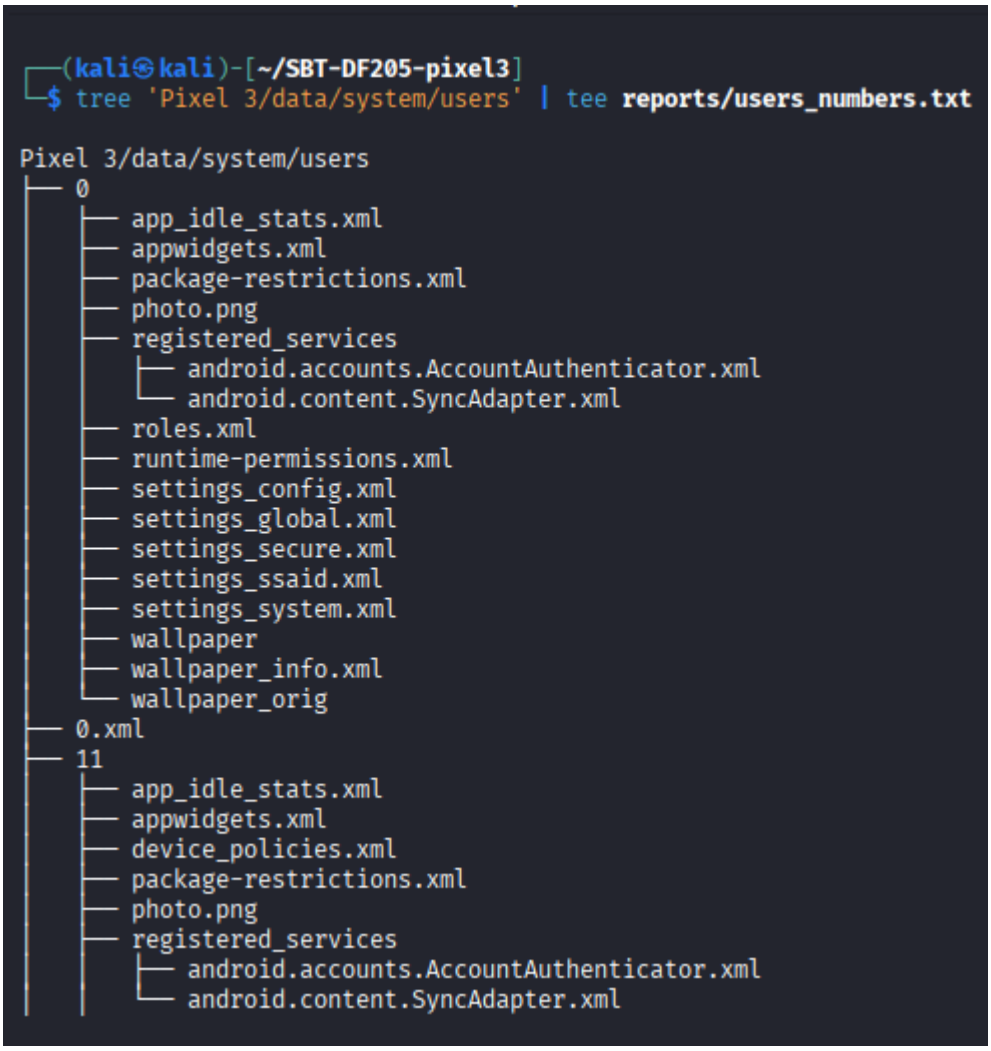


Figure 9. Pixel 3/data/user_de/0/com.android.phone/ (java naming convention)

4. **Account Information**
   Checking the amounts of accounts on the device


┌──(kali㉿kali)-[~/SBT-DF205-pixel3]
└─$ tree 'Pixel 3/data/system/users' | tee reports/users_numbers.txt
Pixel 3/data/system/users
├── 0
│   ├── app_idle_stats.xml
│   ├── appwidgets.xml
│   ├── package-restrictions.xml
│   ├── photo.png
│   ├── registered_services
│   │   ├── android.accounts.AccountAuthenticator.xml
│   │   └── android.content.SyncAdapter.xml
│   ├── roles.xml
│   ├── runtime-permissions.xml
│   ├── settings_config.xml
│   ├── settings_global.xml
│   ├── settings_secure.xml
│   ├── settings_ssaid.xml
│   ├── settings_system.xml
│   ├── wallpaper
│   ├── wallpaper_info.xml
│   └── wallpaper_orig
├── 0.xml
├── 11
│   ├── app_idle_stats.xml
│   ├── appwidgets.xml
│   ├── device_policies.xml
│   ├── package-restrictions.xml
│   ├── photo.png
│   ├── registered_services
│   │   ├── android.accounts.AccountAuthenticator.xml
│   │   └── android.content.SyncAdapter.xml

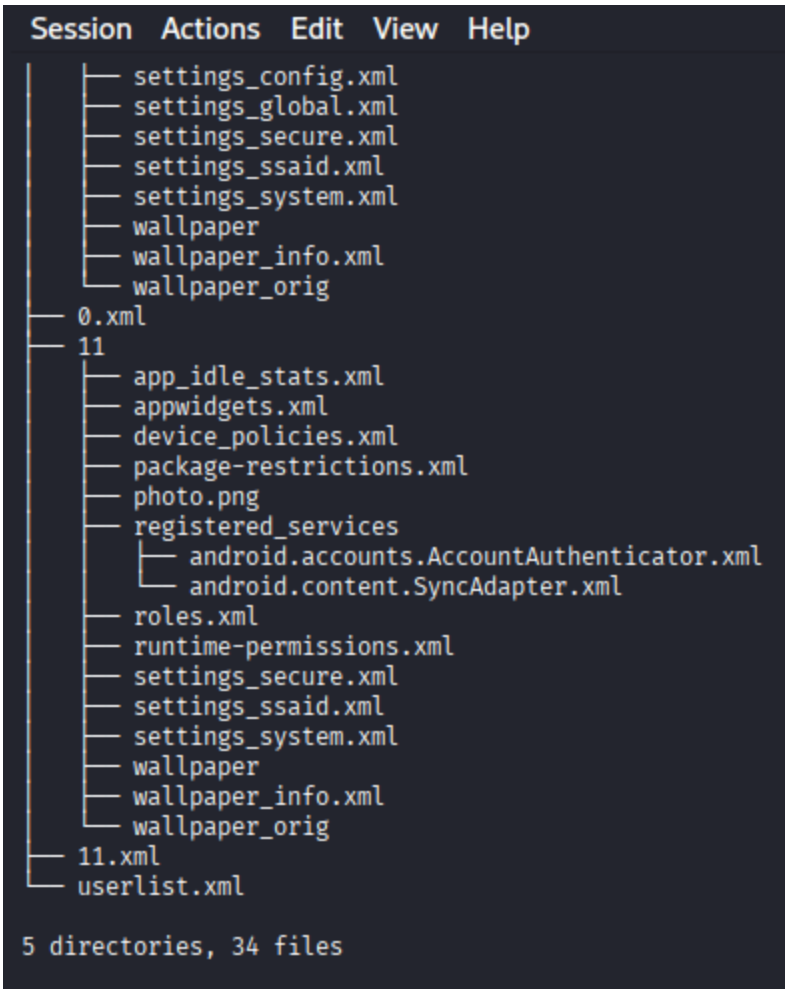


Figure 10. E07: The structure shows (users) meaning multiple users are on the device. the 0 user is the owner/primary user totally consisting of 5 directories, 34 files. The userlist.xml file stores all the users information on the device.

cat 'Pixel 3/data/system/users/0.xml' (this was used to extract the last time the owner or user logged in)
cat 'Pixel 3/data/system/users/11.xml'



Figure 11. E08: the owner and other users last time login time captured successfully and converted to UTC.

5. Extracting Email Address on the device

5.1 Counting the number of all email addresses on the pixel 3 device

```
┌──(kali㉿kali)-[~/SBT-DF205-pixel3]
└─$ grep -EiorhI '([[:alnum:]_.-]+@[[:alnum:]_.-]+?\.[[:alpha:].]{2,6})' 'Pixel 3'| sort|uniq -c | sort -nr | wc -l

618
```

Top 20 number of emails on the device

```
┌──(kali㉿kali)-[~/SBT-DF205-pixel3]
└─$ grep -EiorhI '([[:alnum:]_.-]+@[[:alnum:]_.-]+?\.[[:alpha:].]{2,6})' 'Pixel 3'| sort|uniq -c | sort -nr | head  -n 20 | tee reports/email_numbers.txt

   608 thisisdfir@gmail.com
   176 extension-devs@eff.org
   120 thisisdfirtwo@gmail.com
    96 19842352054_1oHEUWGxDxo499MnzUJStpXDO1YoMyeXX4O6Faqv4Acw@prod.tncp.textnow.com
    40 android.hardware.graphics.mapper@2.1.so
    40 android.hardware.graphics.mapper@2.0.so
    40 android.hardware.graphics.common@1.1.so
    40 android.hardware.graphics.common@1.0.so
    38 support@enflick.com
    38 android.hardware.renderscript@1.0.so
    38 19842032223@prod.tncp.textnow.com
    32 ff7a7f86a2e8481abacfb5e046d5046a@patrol-sg.snssdk.com
    24 android.hidl.safe_union@1.0.so
    24 android.hidl.memory.token@1.0.so
    24 android.hidl.memory@1.0.so
    24 android.hardware.graphics.mapper@3.0.so
    24 android.hardware.graphics.common@1.2.so
    20 libstagefright_bufferpool@2.0.1.so
    18 android.system.suspend@1.0.so
    18 android.hidl.token@1.0-utils.so
```

Figure 12. E09. The total number of the emails on this device is 618, first 20 emails on the device are listed above.

5.2 finding which email address is registered for Google services

Examining all generated data/config of Gmail

```
┌──(kali㉿kali)-[~/SBT-DF205-pixel3]
└─$ tree 'Pixel 3/data/data/com.google.android.gm' -L 1 | tee reports/googleservices_emails

Pixel 3/data/data/com.google.android.gm
├── app_webview
├── cache
├── databases
├── files
├── no_backup
└── shared_prefs

7 directories, 0 files
```

Figure 13. All above are the Gmail package, preference often is associated with Gmail info

5.3 Showing shared preferences/configuration

```
┌──(kali㉿kali)-[~/SBT-DF205-pixel3]
└─$ tree 'Pixel 3/data/data/com.google.android.gm/shared_prefs' -L 1 | tee reports/shared_preference

Pixel 3/data/data/com.google.android.gm/shared_prefs
├── ACCOUNT_COUNT_MAP.xml
├── Account-thisisdfir@gmail.com.xml
├── AndroidMail.Main.xml
├── ANIMATABLE_LOGO_VIEW_FILE_KEY.xml
├── attachmentManager_-1294433372.xml
├── c9edm_-1294433372.xml
├── com.google.android.apps.gmail.accountDumpState&#58;thisisdfir@gmail.com&#58;com.google.xml
├── com.google.android.apps.gmail.dumpState.xml
├── com.google.android.apps.gmail.notifications&#58;thisisdfir@gmail.com.xml
├── com.google.android.gm_preferences.xml
├── com.google.android.gms.appid.xml
├── COMPOSE_UPLOADERS.xml
├── FirebaseAppHeartBeat.xml
├── FlagPrefs.xml
├── G6yPrefs.xml
├── GigSyncEngine_thisisdfir@gmail.com.xml
├── GMAIL_NETWORK_LOGGING_DEPOT.xml
├── Gmail.xml
├── _has_set_default_values.xml
├── MailAppProvider.xml
├── phenotype__com.google.android.libraries.social.populous.xml
├── timeZoneDbVersion.xml
├── triaged_notification_receive_times_thisisdfir@gmail.com.xml
├── UnifiedEmail.xml
├── warm_accounts_preferences.xml
├── watermark_notification_tokens_thisisdfir@gmail.com.xml
└── WebViewChromiumPrefs.xml

1 directory, 27 files
```

Figure 14 (E10). Showing the shared configuration with preferences

5.4 Checking Gmail preference setting (cat 'Pixel 3/data/data/com.google.android.gm/shared_prefs/Gmail.xml')

```
┌──(kali㉿kali)-[~/SBT-DF205-pixel3]
└─$ cat  'Pixel 3/data/data/com.google.android.gm/shared_prefs/Gmail.xml' | tee reports/prefs_settings

<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<map>
    <string name="thisisdfir-account-alias">thisisdfir</string>
    <string name="858233690-account-alias">858233690</string>
    <string name="TextNow-account-alias">TextNow</string>
    <string name="active-account">thisisdfir@gmail.com</string>
    <string name="removal-action">archive</string>
    <boolean name="conversation-overview-mode" value="true" />
    <string name="Signal-account-alias">Signal</string>
    <boolean name="conversation-list-swipe" value="true" />
    <boolean name="force_show_welcome_tour" value="false" />
    <boolean name="confirm-archive" value="false" />
    <boolean name="confirm-send" value="false" />
    <string name="thisisdfir@gmail.com-account-alias">thisisdfir@gmail.com</string>
    <string name="Skype-account-alias">Skype</string>
    <string name="+19195794674-account-alias">+19195794674</string>
    <string name="WhatsApp-account-alias">WhatsApp</string>
    <boolean name="confirm-delete" value="false" />
    <int name="welcome_tour_version" value="1" />
    <boolean name="mail-enable-threading" value="true" />
    <string name="cache-google-accounts-synced">thisisdfir@gmail.com</string>
</map>
```

Figure 15 (E11). References and configuration settings showing Strings: TextNow which tells that this Gmail is associated with

a. Phone

b. Apps: skyper, signal, whatsapp

5.5 checking which email is registered in App Store

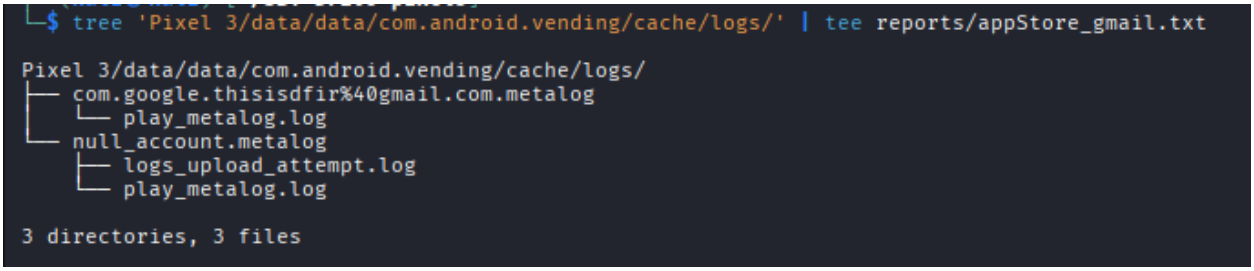


Figure 16. E11. Gmail registered with App Store was captured

5.6 Capturing the Findings of the emails that are associated with scheduled jobs

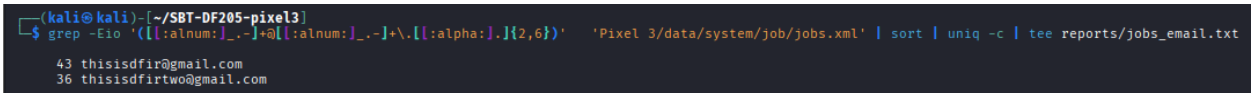


Figure 17. E12 Gmail associated with job scheduling were found.

Conclusion

The archive was hashed before analysis (SHA-256 ca6918ef…8aa00a) and the working copy matched, supporting its integrity. The device is a Google Pixel 3 (codename blueline) running Android 10 (SDK 29), build QQ1A.200105.003, with a security patch level of 2020-01-01. Two user profiles exist: owner user 0 (“This Is”) and user 11 (“User 2”). Both last logged in on 14 February 2020, at 15:51:57 and 12:50:01 UTC respectively. Two Google Fi SIM records were found. One carries the number +19195794674, which matches the phone-number alias in the Gmail preferences. This corroborates the link between that line and the account thisisdfir@gmail.com. That account also appears in the Play Store metalog and the scheduled-job records, and a second account, thisisdfirtwo@gmail.com, appears in the job records. One blocked number (+19195121037) was recorded.

These records establish the device, profiles and accounts, but they do not by themselves show unauthorised communication or information

sharing, so the allegation remains unresolved on this evidence. Profile and account records cannot attribute activity to a named individual, and alias strings in Gmail preferences do not prove that the named apps were used. Further evidence needed: SMS/MMS and call records, WhatsApp and TextNow databases, and media and location records, correlated on a UTC timeline.